

Assignment #02

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F24-BSAI-0009

Machine Learning

Topic:

"Linear Regression Intuition"

Equation:-

$$\hat{y} = mx + b$$

Error Equation:-

$$E = \underset{\substack{\uparrow \\ \text{(actual)}}}{y} - \underset{\substack{\uparrow \\ \text{(predicted)}}}{\hat{y}}$$

put $\hat{y}$ in E

$$E = y - (mx + b)$$

Take square

$$\begin{aligned} E^2 &= \frac{1}{n} [y - (mx + b)]^2 \\ &= \frac{1}{n} \sum [y - (mx + b)]^2 \end{aligned}$$

Open square formula

$$\begin{aligned} E^2 &= \sum (y^2 - 2y(mx + b) + (mx + b)^2) \\ &= \sum (y^2 - 2ymx + 2yb + m^2x^2 + 2mbx + b^2) \\ &= \sum y^2 - \sum 2ymx - \sum 2yb + \sum m^2x^2 + \sum 2mbx + \sum b^2 \\ &= \sum y^2 - 2m \sum xy - 2b \sum y + m^2 \sum x^2 + 2mb \sum x + b^2 \quad \text{--- (i)} \end{aligned}$$

Take Partial Derivative of (i)

w.r.t "m".

$$\frac{\partial E^2}{\partial m} = 0 - 2 \sum xy - 0 + 2m \sum x^2 + 2b \sum x + 0$$

$$\frac{\partial E^2}{\partial m} = -2\sum xy + 2m\sum x^2 + 2b\sum x$$

$$\therefore \text{loss} = 0 \Rightarrow \frac{\partial E^2}{\partial m} = 0$$

$$0 = -2\sum xy + 2m\sum x^2 + 2b\sum x$$

Dividing by 2,

$$0 = \frac{-2\sum xy}{2} + \frac{2m\sum x^2}{2} + \frac{2b\sum x}{2}$$

$$0 = -\sum xy + m\sum x^2 + b\sum x$$

$$\sum xy = m\sum x^2 + b\sum x$$

$$\Rightarrow m\sum x^2 + b\sum x = \sum xy \quad \text{--- (i)}$$

Take Partial Derivative of eq(i)

w.r.t. "b",

$$\frac{\partial E^2}{\partial b} = 0 - 0 - 2\sum y + 0 + 2m\sum x + 2b$$

$$\frac{\partial E^2}{\partial b} = -2\sum y + 2m\sum x + 2b$$

$$\therefore \text{loss} = 0, \frac{\partial E^2}{\partial b} = 0$$

$$0 = -2\sum y + 2m\sum x + 2b$$

Dividing by 2,

$$0 = \frac{-2\sum y}{2} + \frac{2m\sum x}{2} + \frac{2b}{2}$$

$$0 = -\sum y + m\sum x + b$$

$$\sum y = m \sum x + b \quad \text{--- (ii)}$$

Put eq (ii) in eq (i) by
deriving b value,

$$b = \sum y - m \sum x \quad \text{--- (iv)}$$

$$m \sum x^2 + (\sum y - m \sum x) \sum x = \sum xy$$

$$m \sum x^2 + \sum xy - m (\sum x)^2 = \sum xy$$

$$m \sum x^2 - m (\sum x)^2 = \sum xy - \sum x \sum y$$

$$m (\sum x^2 - (\sum x)^2) = \sum xy - \sum x \sum y$$

$$\Rightarrow m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

Put m in eq (iv)

$$nb = \sum y - \left(\frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \right) \sum x$$

$$= \sum y - \frac{n \sum x \sum xy + (\sum x)^2 \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{\sum y (n \sum x^2 - (\sum x)^2) - n \sum x \sum xy + (\sum x)^2 \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{n \sum x^2 \sum y - (\sum x)^2 \sum y - n \sum x \sum xy + (\sum x)^2 \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{n (\sum x^2 \sum y - \sum x \sum xy)}{n (\sum x^2 - (\sum x)^2)}$$

Day: _____

Date: _____

$$\Rightarrow b = \frac{\sum x^2 \sum y - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$$